Solution Design Document — Document Management System (DMS)

Oct 30, 2026 · School ERP, Pamantasan ng Lungsod ng Pasig · Version 2.0 (draft)

1. Document control

Version history

| Version | Date | Author | Change summary |
|---|---|---|---|
| 0.1 | 2026-10-10 | Thomas Dominique Baran | Architecture diagrams drafted (5 pages) |
| 0.2 | 2026-10-13 | Thomas Dominique Baran | Software design document drafted; scope set as the ERP document repository; students out of scope |
| 0.3 | 2026-10-13 | Thomas Dominique Baran | Restructured into the Solution Design Document format; reviewers and executive summary added |
| 0.4 | 2026-10-13 | Thomas Dominique Baran | Team edits: objectives, scope, FR-03, NFR-03; decisions log AD-01 to AD-07 (local ClamAV scanning is AD-01); architecture figures added |
| 0.5 | 2026-10-18 | Thomas Dominique Baran | Aligned with the 18 Oct implementation plan: Super Admin, Admin and Staff personas; invitation-only accounts; approvals; workspace registry; OAS-to-ES packet release; delivery status updated |
| 0.6 | 2026-10-21 | Thomas Dominique Baran | Availability, recovery, audit retention, accessibility and capacity targets filled; persona estimates added |
| 1.0 | 2026-10-21 | Thomas Dominique Baran | Version history updated; proposed owners named for migration tasks |
| 1.1 | 2026-10-23 | Thomas Dominique Baran | Scope reduction: Removed PMS, EMS, and ES document custody. DMS is now strictly the OAS-to-ES bridge and internal workspace. Dropped FR-11 (Memos). |
| 2.0 | 2026-10-30 | DMS team | Revised hosting and support responsibilities, expanded logical flow, clarified implemented and pending security controls, and replaced |

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| | | | testing matrix with compact release evidence; Arial 11 and justified layout. |

Reviewers and approvers

| Name | Role | Organisation | Responsibility |
|---|---|---|---|
| Joseph Wilfred Dela Cruz | Client | Pamantasan ng Lungsod ng Pasig (Client) | Reviewer: Client Requirements |
| Niccolo Yumul | Enterprise Architect | Malunggay Pandesal (Employee Management System) | Approver: Enterprise Solutions |
| Prespurt Bellen | Solutions Architect | Malunggay Pandesal (Employee Management System) | Reviewer: EMS Solution |
| Nathan Fernandez | Solutions Architect | Antonio Pigafetta (Payroll Management System) | Reviewer: PMS Solution |
| Thomas Baran | Solutions Architect | Nagpayong Pinagbuhatan (Document Management System) | Reviewer: DMS Solution |
| Neil Aguilar | Solutions Architect | (Enrollment System) | Reviewer: ES Solution |
| Gabriel Abella | Solutions Architect | Avisala (Online Admission System) | Reviewer: OAS Solution |

Related documents: Architecture diagrams (ARCHITECTURE.drawio); Business Requirements Document, DPIA, Test Strategy and Cutover Plan.

2. Executive summary

The school ERP for Pamantasan ng Lungsod ng Pasig is made of four independent systems: Employee Management (EMS), Payroll Management (PMS), Enrollment (ES) and Online Admission (OAS).

There is currently no secure, controlled place to hold cross-system documents. We propose the Document Management System (DMS) as a centralized repository and secure bridge. It is one Laravel 12 application with its own MySQL database and private file storage.

The primary cross-system use case is the secure transfer of verified applicant documents from OAS to ES, eliminating the need for students to resubmit requirements. The four systems also send transactional history to the DMS through a scoped, idempotent API for centralized auditing. Staff use a web workspace to store, share, and audit internal documents. Every file is validated, scanned, and checksummed. The design needs no managed cloud services, keeping running costs low. Students are not DMS users.

3. Business context, objectives and scope

3.1 Organisation profile: The client is Pamantasan ng Lungsod ng Pasig, Pasig City, Philippines.

3.2 Business drivers:

- Four ERP systems keep separate databases; critical cross-system documents (like admission requirements) are not held in one controlled place.
- Evidence must be preserved: earlier file versions, received history, and audit entries cannot be overwritten.
- Integration must tolerate retries and out-of-order updates without duplicating or corrupting records.
- Running cost must stay low, so the design avoids managed database and storage services.

3.3 Objectives and success measures

| ID | Objective | Measure | Target |
|---|---|---|---|
| OBJ-01 | Centralized ERP document and history repository | Percentage of integrated ERP systems successfully sending documents and transaction history to the DMS | 100% |
| OBJ-02 | Safe retries and duplicate prevention | Number of duplicate history records created when an event is retried or resent | 0 |
| OBJ-03 | Secure and trusted file storage | Number of files served without passing malware scanning or checksum verification | 0 |
| OBJ-04 | Low operational and maintenance cost | Number of additional database engines required beyond the existing MySQL database | 0 |
| OBJ-05 | Complete audit trail and accountability | Percentage of sensitive actions (uploads, downloads, modifications, deletions, and access changes) recorded in the audit log | 100% |

3.4 Scope

| In scope | Out of scope |
|---|---|

| | |
|---|---|
| Document workspace with nested folders, version history, trash, restoration, and safe file previews | File preview support for formats outside of the agreed list |
| Document sharing with named recipients and comments, restricted to authorized users within the organization | External sharing, anonymous or public sharing links |
| Invitation-only account administration, approval of sensitive changes, and audit trail | Business approval or verification of admissions, enrollment, employment or payroll |
| /api/v1 boundary for history, documents, files and events | Source-side adapters and outbox in each ERP system |
| Integration with EMS, PMS, ES, and OAS for receiving transaction history and events | Permanent archival or custody of PMS (payroll), EMS (HR), and ES (enrollment) transactional documents |

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|---|---|
| Storage of received transaction history and the latest snapshot of each source record | Direct management or modification of source records within ERP systems |
| DMS MySQL database and private version-file storage | Folder-level sharing and in-browser editing of document contents |
| Document custody and secure exchange of verified applicant documents from OAS to ES | Modification of business logic and databases within the ERP systems |
| Database-driven workspace registry for ERP and internal workspaces | Students and applicants: not DMS users, no DMS accounts |

3.5 Justification for Centralized Document Custody

The Document Management System (DMS) serves as the centralized repository for institutional documents that require secure preservation, controlled access, version tracking, and authorized exchange between the school's ERP systems.

Although the ERPs maintain their own databases, their responsibilities primarily involve processing business transactions. The DMS provides a dedicated service for preserving the actual documents associated with those transactions, specifically when those documents must cross system boundaries.

The DMS is necessary for the following purposes:

1. Cross-system document exchange: Allow authorized ERP systems (specifically OAS and ES) to retrieve specific verified documents without requiring users to resubmit or manually transfer files.
2. Document integrity: Preserve immutable versions and SHA-256 checksums to verify that official documents have not been altered.
3. Controlled confidentiality: Restrict document access according to classification, originating workspace, and explicitly authorized recipients.
4. Auditability: Record document uploads, retrievals, modifications, releases, and access changes.
5. Reduced document duplication: Allow ERP systems to maintain DMS document identifiers and references instead of separately storing permanent copies of the same files.

Confidential documents shall only be transferred when there is a documented operational, archival, or cross-system requirement. The DMS shall not automatically ingest all records or attachments maintained by an ERP system.

3.6 ERP Document Custody and Business Justification

The DMS does not act as a duplicate permanent archive for every file generated by the ERP systems. Payroll (PMS), Human Resources (EMS), and Enrollment (ES) transactions remain fully within their respective databases. The DMS only holds custody of documents when there is a documented cross-system requirement or an independent institutional preservation need.

The OAS manages applicant submissions and verification. Documents submitted during admission are required by the ES when an admitted applicant proceeds to enrollment.

The DMS preserves the verified document versions and enables authorized retrieval by ES without requiring the applicant to submit the same evidence again. This eliminates manual file transfers and ensures the student's data is securely passed between systems.

Document exchange: OAS → DMS → ES

Only documents explicitly included in an authorized applicant packet shall be released to ES. Medical records are excluded from the ordinary OAS-to-ES exchange.

Proposed document inventory:

| Document Type | Purpose | DMS Justification | Classification |
|---|---|---|---|
| PSA Birth Certificate | Identity and birth information verification | Preserve verified identity evidence for authorized enrollment retrieval | Confidential |

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|---|---|---|---|
| Form 137 / Form 138 | Academic record verification | Reuse verified academic evidence during enrollment | Confidential |
| Applicant Identification Document | Applicant identity verification | Preserve submitted identity evidence and revisions | Confidential |
| Parent/Guardian Identification Document | Parent or guardian verification | Retain required supporting evidence | Confidential |
| Guardianship Affidavit | Verification of legal guardianship | Preserve applicable guardianship evidence | Confidential |
| Applicant Photograph | Applicant identification | Preserve the submitted photograph associated with the application | Confidential |
| Signed Examination Ticket | Evidence of examination processing | Preserve the signed examination document where applicable | Confidential |

3.6.2 Internal Office Workspaces (Non-ERP Documents)

While the four ERPs handle transactional records, school offices (e.g., Office of the Registrar, Dean's Office, Administrative Office) generate general institutional files that do not belong to a specific ERP database.

The DMS provides secure, access-controlled workspaces for these files, ensuring version history and audit trails are maintained without cluttering the transactional systems.

| Document Type | Purpose | DMS Justification | Classification |
|---|---|---|---|

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|---|---|---|---|
| Institutional Memorandums | Internal policy circulation | Centralized dissemination and audit of released versions | Internal |
| Meeting Minutes / Agendas | Record of official meetings | Immutable record of decisions and attendance | Internal |
| Policy Drafts and Approvals | Document versioning for institutional policies | Track revisions and final approvals | Internal |
| General Office Correspondence | Letters, requests, and external communications | Maintain an auditable trail of communications | Internal |

4. Stakeholders and user personas

User counts are planning estimates, to be confirmed by the client. Peak load follows the system baseline of 100 concurrent browser sessions and 20 concurrent API writers. Applicants and students use their own ERP systems and never receive DMS accounts.

| Persona | Approx. users (estimate) | Key tasks | Primary channel | Peak load |
|---|---|---|---|---|

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| Super Admin (DMS system administrator) | 2 | Invite Admins; manage the workspace registry and connectors; oversee and alter DMS-held documents within the school and campus; approve Admin sensitive changes; review audit | Web | Within the 100-session baseline |
| Admin (office or college repository administrator) | 5 | Manage content and Staff within an assigned workspace; propose Staff invitations; approve Staff sensitive changes | Web | Within the 100-session baseline |
| Staff (faculty, college secretaries, department heads, deans and other school staff) | 300 | Keep own files; read shared memos; collaborate on assigned documents; request sensitive changes | Web | Within the 100-session baseline |
| ERP Representative (Staff in Registrar/Admission, HR and Payroll, affiliated to ES, OAS, EMS or PMS) | 20 | Store and review documents in their ERP workspace; open documents shared with them | Web | Within the 100-session baseline |
| ERP System Connector | 4 systems | Send history and documents; read back history and events | API | 20 concurrent writers; 120 requests per minute per connector |

5. Requirements

Priorities follow MoSCoW (Must, Should, Could, Won't) and are proposed for client confirmation. A full list is held in the Business Requirements Document.

5.1 Functional requirements

| ID | Module | Requirement | Priority |
|---|---|---|---|
| FR-01 | Documents | Upload files as immutable versions after content validation and malware scan; keep every earlier version | Must |
| FR-02 | Workspace | Nested folders, rename, move, star, trash and restore, safe previews, grid and list views | Must |
| FR-03 | Sharing | Document sharing grants Viewer, Commenter or Editor permissions, with expiry of at most 30 days and revocation. Editors can rename the document and upload new versions (revision-checked). Editors cannot move, delete or reshare. | Must |
| FR-04 | Comments | Comments on shared documents (up to 2,000 characters) and an activity sidebar | Should |
| FR-05 | Accounts | Invitation-only accounts: Super Admin invites Admins; Admin proposes Staff within its workspace and Super Admin approves; the recipient sets their own password; Staff cannot invite; only Super Admin grants Admin or Super Admin privileges | Must |
| FR-06 | History intake | Accept history events from EMS, PMS, ES and OAS; store an immutable receipt; update the latest snapshot only for newer revisions | Must |

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|---|---|---|---|
| FR-07 | History read-back | Cursor-based reads of history and events through the API; office and admin UI to inspect received versus current state | Must |
| FR-08 | Audit | Record actor, action and document for sensitive actions | Must |
| FR-09 | Scoped access | Limit users and connectors to their school, campus, workspace and, for Admin and Staff, their assigned memberships | Must |
| FR-10 | Workspace registry | Database-driven registry of ERP and internal workspaces, each with a stable code, editable name and lifecycle state; a permission-filtered selector; adding or renaming a workspace needs no code change | Must |
| FR-11 | Applicant packet | OAS releases specific verified document versions of an admitted applicant; ES retrieves only those Clean versions through an explicit grant; no medical files through the ordinary connector. | Must |
| FR-12 | Approvals | Sensitive changes (sharing, classification, trash or restore, account changes): Staff requests go to a scoped Admin and Admin requests to Super Admin; approve or reject with a reason; no self-approval. | Must |
| FR-13 | Document Type Registry | Maintain a database-driven registry of document types, originating workspaces, classifications, permitted formats, and authorized custody policies. | Must |

| | | | |
|---|---|---|---|
| FR-14 | ERP Document Intake | Validate incoming documents against their registered type, authenticated originating workspace, permitted format, and required metadata before accepting them. | Must |
| FR-15 | Controlled Document Exchange | Permit cross-ERP document retrieval only through an explicit, scoped, auditable release grant referencing authorized document versions. | Must |
| FR-16 | Document Custody | Preserve immutable official documents under DMS custody while retaining business-record ownership in the originating ERP. | Must |
| ID | Module | Requirement | Priority |

5.2 Non-functional requirements

| ID | Category | Requirement |
|---|---|---|
| NFR-01 | Security | Session sign-in with active-account check, CSRF protection and sign-in throttling; connector tokens stored hashed and bound to source, school and campus; HTTPS with TLS 1.2 or later and secure cookies; least privilege |
| NFR-02 | Integrity | SHA-256 checksum recorded for every version and re-verified on access; versions are never changed |
| NFR-03 | Idempotency | A retried request with the same key and content returns the original receipt (200). A changed request or stale revision returns 409. An older revision is stored (applied=false) without changing the latest snapshot. |

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|---|---|---|
| NFR-04 | Limits | API throttle 120 requests per minute per connector; file up to 10 MB; history payload up to 256 KiB; folder upload up to 50 files per batch; capacity baseline of 100 concurrent browser sessions and 20 concurrent API writers, to be validated by a load test on the target host |
| NFR-05 | Availability | 99.5% monthly service availability, excluding announced maintenance, measured by authenticated health checks |
| NFR-06 | Recovery | RPO 24 hours and RTO 8 hours for single-site recovery; daily encrypted backup and quarterly restore drill; MySQL and private files restored together |
| NFR-07 | Privacy | Compliant with the Philippine Data Privacy Act of 2012 (RA 10173); data residency to be agreed with the client and the Enterprise Architect |
| NFR-08 | Auditability | Audit history kept for three years, online or archived; a legal hold suspends disposal |
| NFR-09 | Accessibility | WCAG 2.2 Level AA target for staff-facing pages, with keyboard navigation and a skip link |

6. Current state and gap analysis

| Current system | Function | Data held | Pain point / gap | Disposition |
|---|---|---|---|---|
| Employee Management System (EMS) | Employee management | Own database (outside DMS) | Documents and change history not held in one controlled place; source adapter not yet built | Retain and integrate |
| Payroll Management System (PMS) | Payroll management | Own database (outside DMS) | As above | Retain and integrate |
| Enrollment System (ES) | Enrollment | Own database (outside DMS) | As above | Retain and integrate |

| | | | | |
|---|---|---|---|---|
| Online Admission System (OAS) | Online admission | Own database (outside DMS) | As above | Retain and integrate |

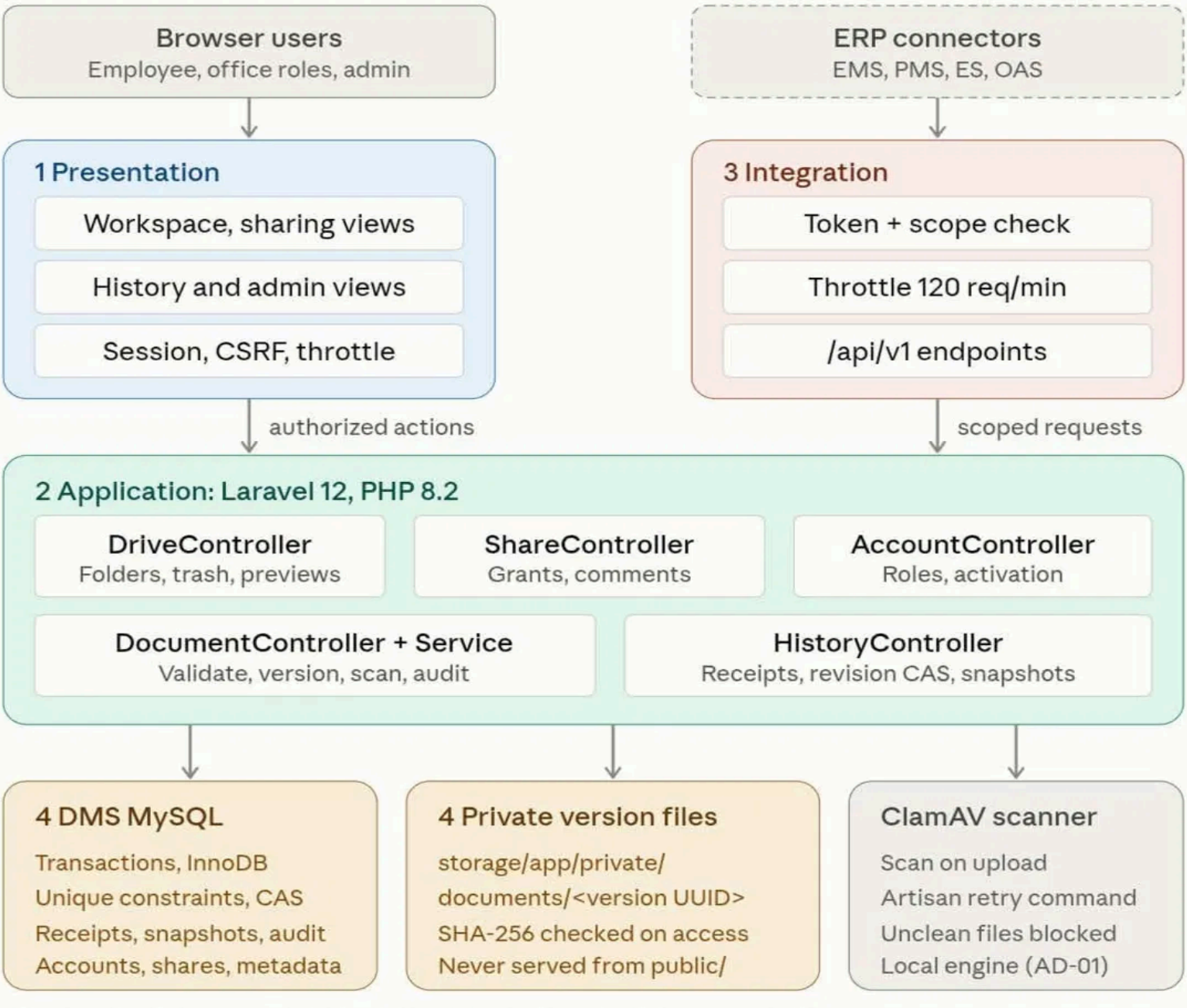
Source-side adapters are planned work for each ERP team. Data-quality checks, source identity matching and per-source payload schemas have not been done yet.

7. Target solution architecture

7.1 Architecture principles

- Clear Data Ownership: ERP systems manage their respective business records. The DMS manages cross-system documents, versions, sharing, audit logs, and received history.
- Unified Application Architecture: The DMS uses a single Laravel application organized into four logical tiers, without requiring separate servers or microservices.
- Controlled System Integration: All ERP systems communicate with the DMS through the /api/v1 endpoints, without direct access to each other's databases.
- Secure File Storage: Files are stored in private directories outside the publicly accessible web folder, with checksums used to verify file integrity.
- Reliable Data Processing: The system prevents duplicate records and handles simultaneous requests through request receipts, revision validation, and database constraints.
- Cost-Effective Infrastructure: The DMS uses a single MySQL database and a straightforward folder-based storage structure to minimize operational costs.

### 7.2 Logical architecture




Browser → Presentation (Blade, session UI) → Application (controllers, policies, services) → Persistence (DMS MySQL + private files)

EMS / PMS / ES / OAS → Integration (/api/v1, scoped tokens) → Application

Application → database queue → supervised ClamAV worker → scan result

Figure 1: Compact logical flow. These are responsibilities inside one Laravel deployment, not separate servers. Browser actions and ERP requests enter through different authenticated

boundaries. The application authorizes and validates each action, coordinates database writes and private files, and records audit and API receipts. ERP systems keep their own operational databases; DMS receives agreed snapshots and events, never writes to those databases. The worker scans committed versions before download or preview; the scheduler retries eligible failures and runs configured backups. MySQL metadata and private bytes must be restored as one consistent set.

7.3 Component responsibilities

Figure 2: Component view. Controllers, throttle, ClamAV scanner and storage paths.

| Component | Responsibility | Traces to |
|---|---|---|
| Presentation (Blade views, workspace) | Sign-in, role-aware navigation, folders, sharing and history screens | FR-02, FR-04, FR-07 |
| Session security | Active account, CSRF, sign-in throttling, school/campus boundaries, grants | NFR-01, FR-09 |
| DocumentController + DocumentService | Content validation, immutable versions, scan, idempotency, checksums, audit | FR-01, FR-08, NFR-02, NFR-03 |
| DriveController | Folders, rename, move, star, trash and restore, previews, share creation and revocation | FR-02, FR-03 |
| ShareController | Viewer, Commenter and Editor grants, pinned downloads, comments | FR-03, FR-04 |
| AccountController | Invitation-only account creation, roles and activation, with approval by the next role up | FR-05 |
| Workspace registry | Registered ERP and internal workspaces, lifecycle, memberships; feeds selectors and assignments | FR-09, FR-10 |
| Approval queue | Pending requests, reviewer routing, authority re-check, atomic apply | FR-13 |
| HistoryController | Canonical receipt hashing, revision ordering, record locks, snapshots | FR-06, FR-07, NFR-03 |
| IntegrationAuth + API routes | Hashed token lookup, source/school/campus binding, throttle, /api/v1 endpoints | INT-01 to INT-05, INT-07, NFR-01, NFR-04 |

| | | |
|---|---|---|
| DMS MySQL database | Transactions, unique constraints, revision compare-and-swap | FR-06, NFR-03 |
| Private storage + scanner | UUID paths, SHA-256, ClamAV scan and retry command | FR-01, NFR-02, INT-06 |

7.4 Build vs buy: All components are custom-built on Laravel 12. Malware scanning uses a local ClamAV engine instead of a paid scanning service (AD-01).

8. Data architecture and data protection

8.1 Core entities and system of record

| Entity | Key attributes | System of record | Consumers |
|---|---|---|---|
| User account | id, email, password hash, role, active, school_id, campus | DMS | Workspace |
| Invitation | recipient email, role, workspace, token hash, status, inviter, approver | DMS | Admins, recipient |
| Workspace and membership | UUID, school/campus, code, name, kind (ERP or internal), lifecycle state; user, workspace, responsibilities | DMS | Selectors, assignments, access checks |
| Folder | UUID, parent_id, owner, name, source, school, campus, revision | DMS | Workspace |
| Document | UUID, owner, folder, title, subject, reference, source, category, classification, revision, trashed_at | DMS | Workspace; ERP systems (read) |
| Document version | UUID, document_id, number, filename, MIME type, size, checksum, storage_key, scan status | DMS (bytes in private storage) | Workspace, shares, ERP systems |
| Share, comment, star | document, version, recipient, permission, expiry, revocation | DMS | Recipients |

| | | | |
|---|---|---|---|
| Release grant | document or exact version, permitted user, group or connector, operation, expiry, revocation, release reference | DMS | ES (released packet), recipients |
| Approval request | requester, target, proposed change, reason, expected revision, reviewer, status | DMS | Approvers |
| Source record snapshot | scoped record key, source, school/campus, record type and ID, revision, deleted, payload | The source ERP system (DMS keeps the latest copy) | Office/admin history UI, API read-back |
| Received history receipt | cursor ID, record key, event ID, revision, request hash, operation, payload, applied | DMS (immutable) | Office/admin history UI, API read-back |
| Audit entry, integration event | school/campus, actor, action, document, event type, revision | DMS | Admins, ERP systems (events) |
| Connector identity, request receipt | source, school/campus, token hash; request key, request hash, response receipt | DMS | API |

8.2 Data classification (proposed; each document carries a classification field and the classes need client sign-off)

| Class | Examples | Controls |
|---|---|---|
| Restricted | Password hashes, token hashes, payroll and HR documents | Source-scoped access, private storage, audit |
| Confidential | Enrollment and admission documents, received history payloads, audit entries | School/campus and source scoping, private storage, checksum |
| Internal | General office working files | Owner and explicit grants |

8.3 Retention and privacy

- Audit history is kept for three years, online or archived, and a legal hold suspends disposal. Retention for documents and received history is agreed with the client, aligned to the applicable records rules.
- A Data Protection Impact Assessment is completed before go-live.
- Data residency: production data and backups are stored in a region decided with the hosting model.

- ES and OAS payloads may contain personal data of students and applicants, who are not DMS users. Treat it as children's personal data and limit history access to office and admin roles.
- Payloads are source evidence, not account-provisioning instructions.
- Subject access and deletion requests: the process is not part of the current design and will be defined separately.

8.4 ERP Document Type Registry

The DMS shall maintain a database-driven document type registry identifying the documents authorized for intake from each ERP workspace.

Each registered document type shall contain:

| Attribute | Description |
|---|---|
| Document Type Code | Unique identifier within the applicable registry scope |
| Document Type Name | Human-readable document name |
| Originating Workspace | ERP or internal workspace responsible for the document |
| Default Classification | Internal, Confidential, or Restricted |
| Permitted File Formats | Allowed file formats for the document type |
| Maximum File Size | File size restriction subject to the DMS system limit |
| Authorized Custodian | Office or role responsible for document custody |
| Authorized Recipients | Users, groups, or connectors eligible for an explicit access grant |
| Cross-System Release | Whether the document type is eligible for an approved ERP exchange |
| Retention Policy Reference | Applicable institutional records schedule |
| Lifecycle Status | Active, disabled for new intake, or retired |

The document type registry shall be independent of the file extension and document classification.

A document type identifies the nature of the institutional record, while its file format identifies the technical representation of its contents.

8.5 Supported File Formats and Validation

The following file formats are proposed for the initial document ingestion policy.

| Format | Extension | Intended Use | Preview |
|---|---|---|---|
| Portable Document Format | .pdf | Official documents, certificates, payslips, signed forms | Supported |
| JPEG Image | .jpg, .jpeg | Scanned identification documents and photographs | Supported |
| PNG Image | .png | Scanned forms, supporting images | Supported |
| Microsoft Word Document | .docx | Authorized office working documents | Download only |
| Microsoft Excel Workbook | .xlsx | Authorized office reports | Download only |

All files shall be subject to the existing DMS controls:

- Maximum file size of 10 MB.
- Actual-content and MIME-type validation.
- Malware scanning through ClamAV.
- Private filesystem storage.
- SHA-256 checksum generation and verification.
- Immutable document versions.
- Access control enforcement before preview or download.

The DMS shall reject files that do not match the permitted formats of their registered document type.

8.6 Confidential Document Intake Policy

The DMS shall not automatically receive every confidential document stored in a connected ERP system.

A document type shall be eligible for DMS custody when at least one of the following conditions is satisfied:

1. The document requires independent institutional preservation.
2. Another authorized ERP system or institutional office requires controlled retrieval.
3. The document requires immutable version tracking and an independent audit history.
4. The institution has designated DMS as the authoritative custodian of the issued document.

Before enabling intake, the document type shall have a defined originating system, authorized custodian, operational purpose, classification, permitted recipients, and applicable retention policy.

Classification alone shall not authorize access. All retrievals shall be evaluated against the user's or connector's effective permissions and the document's explicit access grants.

Medical documents and other specially protected records shall not be accessible through ordinary ERP connector privileges or broad repository membership.

9. Integration architecture

All integrations go through the one /api/v1 boundary. No education data standard is used today; the contract is a JSON format carrying event ID, record identity, operation and revision.

| ID | Interface | Direction | Pattern | Frequency | Data | Standard |
|---|---|---|---|---|---|---|
| INT-01 | EMS | EMS → DMS | REST push, idempotent | Per record change | History events, documents | Bearer token, JSON |
| INT-02 | PMS | PMS → DMS | REST push, idempotent | Per record change | History events, documents | Bearer token, JSON |
| INT-03 | ES | ES → DMS | REST push, idempotent | Per record change | History events, documents | Bearer token, JSON |
| INT-04 | OAS | OAS → DMS | REST push, idempotent | Per record change | History events, documents, applicant packet release | Bearer token, JSON |
| INT-05 | ERP read-back | DMS → ERP (GET) | REST, cursor-based | On demand | History, events, documents, files | Bearer token, JSON |
| INT-06 | Malware scanner | DMS → ClamAV | Command | Per upload and retry | File bytes | ClamAV command, Artisan retry |
| INT-07 | Applicant packet read | DMS → ES (GET) | REST, read-only, explicit release grant | On demand after release | Allowed Clean document IDs and exact version IDs of a released packet | Bearer token, JSON |

Error handling, retries and idempotency:

- New event: 201 with history_id, applied and duplicate. Identical replay: 200 with the original receipt. Same event with a different hash, or another event at the same record revision: 409.

- An older revision is stored with applied=false and the snapshot stays unchanged; a delete becomes a tombstone and earlier payloads are kept.
- Under contention DMS retries a bounded number of times, then returns 409. The source retries the same event with backoff and jitter.
- Throttle: 120 requests per minute per connector. Persistent conflicts need operator investigation; alert recipients are set in the support model.

9.1 ERP Document Exchange Policy

ERP systems shall exchange documents through authenticated /api/v1 interfaces. The DMS shall enforce the following:

1. Each document must identify its originating ERP workspace and registered document type.
2. The originating ERP must supply a stable external record reference.
3. DMS shall preserve the document and its immutable versions.
4. Cross-system access requires an explicit release grant.
5. A release grant shall identify the authorized recipient, permitted operation, and applicable document or exact version.
6. Every retrieval shall revalidate authorization, scan status, and file integrity.
7. All cross-system retrievals shall be recorded in the audit trail.
8. The receiving ERP shall not obtain general access to the originating ERP's document repository.
9. Revocation shall prevent subsequent retrievals without deleting historical evidence.

The initial and primary cross-system document exchange shall be the verified admission documents released from OAS to ES.

10. Security, identity and compliance

10.1 Identity and access

- Staff sign in with school accounts (email and password hash) in a session; inactive accounts are refused.
- Accounts are invitation-only: an individual, expiring, single-use link lets the recipient set their own password. Sensitive changes (sharing, classification, trash or restore, official publication, account changes) by Admin need Super Admin approval and by Staff need Admin approval; no self-approval.
- ERP systems authenticate with a bearer token stored hashed and bound to one source, school and campus.
- Single sign-on and multi-factor authentication are not in the current design; the Enterprise Architect decides whether to add them.

10.2 Role matrix

| Role | Own files | Shared files | Workspace documents | Received history | Accounts and audit |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Super Admin | Read, write (personal workspace) | Per grant | All DMS repositories within school and campus; access audited | Read | Manage; full scoped audit |
| Admin | Read, write | Per grant | Assigned workspace only; sensitive changes need Super Admin approval | Assigned source | Propose Staff invitations; assigned-source audit |
| Staff (including ERP Representatives) | Read, write | Per grant | Owned, assigned or explicitly shared documents; sensitive changes need Admin approval | None | Own request history only |
| ERP connector | n/a | n/a | Own workspace, school, campus (API) | Write and read own | None |

**10.3 Security controls**

Implemented in DMS: active-account and role checks; school, campus and source scoping; CSRF and sign-in/API throttling; hashed connector tokens; private UUID file paths; content, size and MIME checks; SHA-256 verification; ClamAV scan gating; immutable versions, revision/idempotency constraints and audit entries. A blocked or unscanned file is not served. The application can create encrypted coordinated backups when configured.

Deployment and release controls: serve only public/ over valid HTTPS; restrict database, storage, secrets and backups; supervise the worker and scheduler; monitor failures; and verify restore. Encryption of the live database/files, central monitoring and independent penetration testing have no completed design or evidence in this draft. The client and Enterprise Architect must decide and document them before go-live; do not present them as active controls.

10.4 Compliance checklist

- Data protection law: Philippine Data Privacy Act of 2012 (RA 10173)
- Personal data of minors inside ES and OAS payloads assessed in the DPIA
- TLS certificates and HTTPS-only access in place
- Backup and restore of MySQL plus private files tested
- Records retention rules for the client agreed
- Accessibility: WCAG 2.2 Level AA target for staff-facing pages

## 11. Infrastructure, deployment and operations

### 11.1 Hosting and environments

The Enterprise Architect and client must choose the host, data residency, budget and support owner. The design needs one Laravel 12/PHP 8.2 deployment, MySQL 8/InnoDB, private version storage, a supervised database queue worker and scheduler, local ClamAV, outbound mail and an HTTPS endpoint reachable by the four ERP systems. Only public/ is web-served. Development and integration testing use synthetic data; UAT uses anonymised data; production holds live records. Keep each environment's database, files, keys and connector tokens separate. No managed database, queue, search or object store is required by the design.

### 11.2 Operations and support

The operator must supervise the web process, worker and scheduler; apply migrations without reseeding live data; monitor HTTPS, database, queue failures, scan backlog, disk capacity and backup results; and assign an incident contact for persistent API conflicts. The authenticated /admin/health endpoint reports database, scan and queue state but does not prove worker liveness or external availability. ERP adapters must retain unsent events and retry the same event/key after outages.

The application can create a daily encrypted archive of a MySQL dump and private version files when backup settings and the scheduler are enabled. An operator must keep the archive and keys in restricted, separate storage, verify restoration quarterly, and record elapsed recovery time. The 99.5% monthly availability, 24-hour RPO and 8-hour RTO are acceptance targets, not achieved service levels. Support hours, escalation tiers, alert recipients, retention and the final SLA require client/host decisions. Capacity of 100 browser sessions and 20 API writers must be measured on the chosen host.

12. Migration, implementation roadmap and testing

12.1 Delivery status and roadmap. Dates and phases are pending the Enterprise Architect.

| Item | Status | Owner (proposed) |
|---|---|---|
| DMS workspace, document custody, sharing, audit (current baseline) | Implemented | DMS team |
| DMS /api/v1 boundary (history, documents, files, events) | Implemented | DMS team |
| Source adapter and outbox for EMS | Planned | EMS team |
| Source adapter and outbox for PMS | Planned | PMS team |
| Source adapter and outbox for ES | Planned | ES team |
| Source adapter and outbox for OAS | Planned | OAS team |
| Workspace registry and dynamic selector | Planned | DMS team |

| | | |
|---|---|---|
| Invitation-only accounts and sensitive-change approvals | Planned | DMS team |
| OAS-to-ES verified applicant packet release | Planned | OAS, ES and DMS teams |
| Payload schemas and source identity matching | Not started | ERP teams with the DMS team |
| Full import (initial snapshots) | Not started | ERP teams (sending) and DMS team (receiving) |
| Retention schedule | Not started | Client |
| Tested restore | Not started | DMS team |
| Single sign-on | Not started | Enterprise Architect |

12.2 Data migration approach

1. Agree record types and payload schemas with each ERP team.
2. Each source commits its local change and an outbox entry together, then sends a consistent initial snapshot with increasing revisions and a stable event ID.
3. DMS stores each snapshot and receipt; the source reconciles using cursor-based reads of /api/v1/history.
4. Run rehearsal loads and reconcile record counts per source with each reviewer.
5. Cut over per source in a window agreed with the Enterprise Architect.

**12.3 Testing strategy**

Automated checks (DMS team): On each push or pull request, run PHP feature tests on isolated SQLite and MySQL 8.0 databases, frontend checks, asset and formatting checks. MySQL checks cover concurrent receipts, revision conflicts, uploads and restore; SQLite alone does not prove row-lock behaviour. The post-merge smoke test verifies migrations, views, login and unauthenticated API rejection. The real ClamAV check runs separately on a schedule or by request.

Release evidence (DMS and ERP teams): Rehearse source-scoped API replay, out-of-order events, token revocation, blocked downloads, office-role access and OAS-to-ES packet release with synthetic data. On the chosen host, measure 100 browser sessions and 20 API writers, verify a monitored queue worker and HTTPS, and complete a timed restore of the matching MySQL dump and private files. Record results, defects, fixes and reviewer sign-off; the 99.5% availability and 24-hour RPO/8-hour RTO remain targets until measured in operation.

12.4 Training and change management: role-based guides for office staff, and an /api/v1 integration guide for the four ERP teams, written once the contract is final.

13. Risks, assumptions, decisions and appendices

13.1 Risks (ratings proposed)

| ID | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| R-01 | Hosting model not settled | Medium | High | Decide with the Enterprise Architect; keep the design hosting-neutral | Enterprise Architect |
| R-02 | ERP adapters late, or revisions not increasing | Medium | High | Publish the API contract; DMS stores out-of-order events as applied=false; reconcile by cursor | ERP Solutions Architects |
| R-03 | MySQL and files not backed up or restored together | Medium | High | Joint backup and a tested restore | DMS Solutions Architect |
| R-04 | Multi-writer MySQL contention untested | Medium | Medium | Load test; bounded retries; 409 handling | DMS team |
| R-05 | Student or applicant data inside ES/OAS payloads | Medium | Very high | DPIA; restrict history to office and admin roles | Client / DPO |
| R-06 | Scanner unavailable blocks uploads | Medium | Medium | Scan-retry command; monitor pending files | DMS team |
| R-07 | Applicant packet release state is stale or ambiguous | Medium | High | Release by authenticated OAS events with revisions; fail closed; recheck the grant on every read | OAS and ES Solutions Architects |
| R-08 | Medical files in OAS packets not yet confirmed | Medium | High | Confirm with the OAS owner before intake; keep them off the ordinary OAS and ES connector until access rules are agreed | Enterprise Architect |

13.2 Assumptions and constraints

- Each ERP team builds its source adapter and keeps revisions increasing per record.

- DMS users are school staff; students have no access.
- The client and the Enterprise Architect decide hosting, budget and data residency.
- Each ERP solution architect reviews the integration contract for their system.

13.3 Architecture decisions log

| ID | Decision | Options considered | Rationale | Date |
|---|---|---|---|---|
| AD-01 | Use a local ClamAV engine, run through a scan-retry Artisan command, for malware scanning instead of commercial managed scanning APIs | Third-party SaaS (VirusTotal, AWS GuardDuty); ClamAV daemon / CLI | Aligns with the low operational cost requirement (OBJ-04); works on-premise or hosted; avoids external vendor recurring costs | 2026-10-13 |
| AD-02 | One Laravel deployment, custom-built, with four logical tiers | Microservices; separate API service | Small operating footprint; shared DocumentService | 2026-10-13 |
| AD-03 | One MySQL database and private folders | Managed cloud database; object storage | Lowest running cost; one backup set | 2026-10-13 |
| AD-04 | Scoped bearer tokens, idempotent receipts, revision compare-and-swap | Direct database access; shared accounts | No writes to other systems' databases; safe retries | 2026-10-13 |
| AD-05 | File scan is a security control, not business approval | Business verification inside DMS | Approval belongs to school users and originating systems | 2026-10-13 |
| AD-06 | Students are not DMS users | Student accounts and workspaces | Outside the agreed scope | 2026-10-13 |
| AD-07 | Hosting model | Local or on-premises; cloud | Open: pending the Enterprise Architect | Open |
| AD-08 | Workspaces come from a database registry, not a fixed list of four | Fixed four-source list in code | A new ERP or internal workspace appears without code changes | 2026-10-18 |

| | | | | |
|---|---|---|---|---|
| AD-09 | Cross-system access only through explicit release grants | Same-school membership as access; broad connector tokens | Least privilege: ES reads only released versions | 2026-10-18 |
| AD-10 | Invitation-only accounts, approved by the next role up | Self-registration; administrator-set passwords | Staff-only environment; the recipient owns their password | 2026-10-18 |

13.4 Appendices

A. Glossary

| Term | Meaning |
|---|---|
| DMS | Document Management System |
| EMS / PMS / ES / OAS | Employee Management, Payroll Management, Enrollment and Online Admission systems |
| Source | One of the four ERP systems when it sends history to DMS |
| Workspace | A registered repository for an ERP system or an internal office |
| Release grant | An explicit permission to read a document or an exact version, given to a named recipient or connector |
| Receipt | Stored result of an API request, returned again on an identical retry |
| Revision | Per-record number from the source that only increases |
| CAS | Compare-and-swap: update only if the revision is newer |
| DPIA | Data Protection Impact Assessment |
| RPO / RTO | Recovery point and recovery time objectives |

B. Requirements traceability (requirement to component to test)

| Requirement | Component | Test |
|---|---|---|
| FR-01, NFR-02 | DocumentService, private storage, scanner | Unit and integration; security |
| FR-02, FR-03, FR-04 | DriveController, ShareController | Acceptance |

| | | |
|---|---|---|
| FR-05, FR-09, NFR-01 | AccountController, invitations, session security, IntegrationAuth | Security |
| FR-06, FR-07, NFR-03 | HistoryController, MySQL | Unit and integration; concurrency |
| FR-08 | DocumentService, audit entries | Acceptance |
| FR-10 | Workspace registry, MySQL | Unit and integration; acceptance |
| FR-11 | DocumentService, ShareController | Acceptance |
| FR-12 | IntegrationAuth, HistoryController, release grants | Security; acceptance |
| FR-13 | Approval queue, AccountController | Security; acceptance |
| NFR-06 | Backup of MySQL and private files | Backup and restore |